10.Python program to implement the Travelling Salesman Problem (TSP)

Aim: To find the minimum-cost tour that visits every city exactly once and returns to the starting city.

Algorithm:

1. Define the distance between each pair of cities.
2. Fix the first city as the starting point.
3. Generate all possible orders of the remaining cities.
4. Calculate the total cost of each tour.
5. Display the tour with the minimum cost.

Program:

python 10.py - C:/Users/praji/AppData/Local/Programs/Python/Python314/python 10.py (3.14.8)

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from itertools import permutations

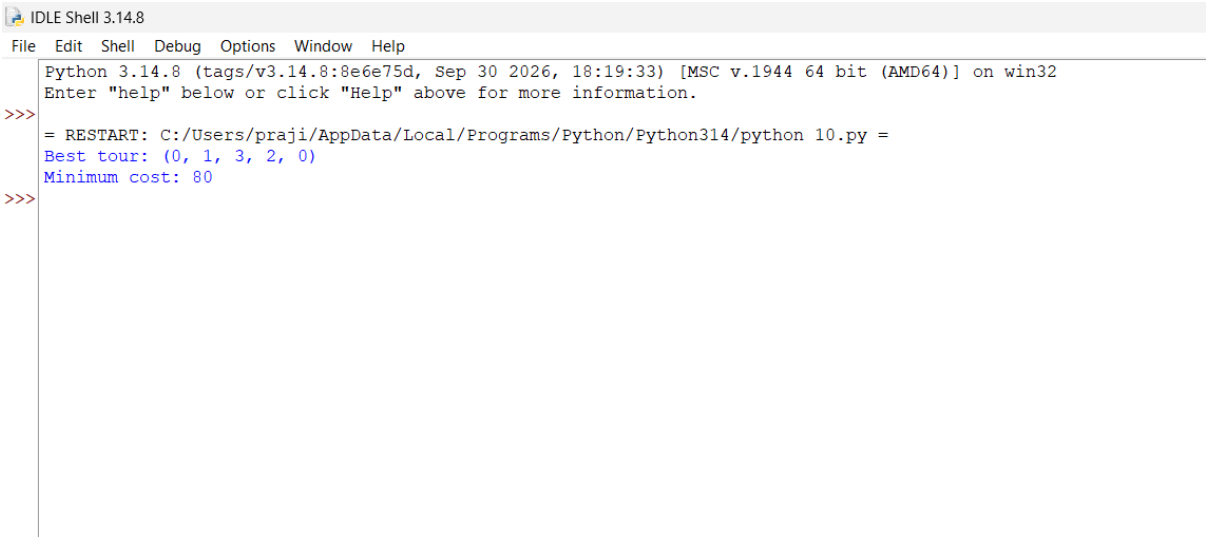
dist = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(dist)
cities = list(range(1, n))
min_cost = float('inf')
best_path = []

for path in permutations(cities):
    tour = (0,) + path + (0,)
    cost = sum(
        dist[tour[i]][tour[i + 1]]
        for i in range(n)
    )

    if cost < min_cost:
        min_cost = cost
        best_path = tour

print("Best tour:", best_path)
print("Minimum cost:", min_cost)
```



Result: Thus, the Travelling Salesman Problem was implemented, and the minimum-cost tour was obtained.